

Kaiyue Chen

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EDUCATION

BEng in Mechatronics and Robotic Systems

Xi'an Jiaotong-Liverpool University, Suzhou, China

Sep 2024 – Jul 2028 (expected)

- GPA (Years 1-2): 3.83/4.0
- Core coursework: Linear Algebra(4.0), Calculus(4.0), C Programming(4.0).

PUBLICATIONS

*Equal contribution.

- 3WD-VLA: Hierarchical Three-Way Decision for Collaborative Dual-View Robotic Grasping.**
Submitted to EAI CollaborateCom 2026, 22nd EAI International Conference on Collaborative Computing: Networking, Applications and Worksharing.
Yan Zhu*, **Kaiyue Chen***, Lingyu Li, Xiaohui Zhu, and Xinheng Wang.
- Real-Time 3-DoF Robotic Grasp Detection via Keypoint Regression.**
Submitted to 2026 31st International Conference on Automation and Computing (ICAC).
Xiaoyu Xia, Chengcheng Hu, Leyi Liu, Junyan He, **Kaiyue Chen**, Yuhao Jin, Yong Yue, and Xiaohui Zhu.

RESEARCH EXPERIENCE

Tactile-Guided VLA for Contact-Rich Robotic Manipulation, Research Intern

Jun 2026 – Present

Sudo Robotics, advised by Prof. Rui Chen

- Designed and deployed vision-based tactile sensors on bimanual grippers to mitigate partial observability caused by occlusion and limited visual feedback.
- Built and iterated on a dual-arm tactile-VLA pipeline based on SmolVLA (0.7B).
- Developing a tactile video encoder to capture contact geometry, shear deformation, and slip-related dynamics from optical tactile observations.
- Exploring Flow Reversal Steering to map tactile-guided coarse action corrections into the latent noise space of pretrained VLA policies.

ManiSkill-ViTac Challenge 2026, Student Organizer

Mar 2026 – Jun 2026

CVPR 2026 Embodied AI Workshop: real-world language-guided bimanual vision-tactile manipulation

- Built an end-to-end VLA pipeline around UMI-based data collection, covering hardware maintenance, data processing and filtering, model training, and real-robot deployment.
- Integrated AnyTouch tactile representations into a $\pi_{0.5}$ -based VLA using modality-weighted training, establishing the competition baseline
- Designed two long-horizon, contact-rich tasks for model training and online deployment.
- Co-organized the challenge and contributed to the accompanying technical report.

Three-Way Decision VLA, Co-first Author

Jan 2026 – May 2026

Xi'an Jiaotong-Liverpool University, advised by Prof. Xinheng Wang

- Developed a collaborative robotic grasping framework for cases where direct execution is unsafe or unreliable, including ambiguous language commands, target absence, human-hand intrusion, occlusion, and stacked objects.
- Designed a three-way decision framework for VLA policies to defer actions in ambiguous scenarios.
- Built a teleoperation data-collection pipeline for the SO-100 robot arm and trained ACT, $\pi_{0.5}$, and SmolVLA models.
- Improved performance by 27.5 percentage points over a matched SmolVLA baseline without 3WD and demonstrated clearer behavior under ambiguity, occlusion, and object stacking.

COMPETITIONS

National Finals Team Member, Baidu Service Robot Competition

Aug 2025

The 27th China Robot and Artificial Intelligence Competition, Suzhou

Team Member, coached by Prof. Eng Gee Lim (IEEE Fellow)

- Developed a ROS-based autonomous restaurant-service robot with order-driven task coordination, integrating touch-screen interaction, multi-point navigation, food pickup, delivery, and checkout.

TECHNICAL SKILLS

Robotics: PyTorch, Deep reinforcement learning, ROS1/2, Gazebo, Isaac Sim, ManiSkill.

Programming: Python, C, C++.

Design and tools: SolidWorks, Autodesk Fusion 360.